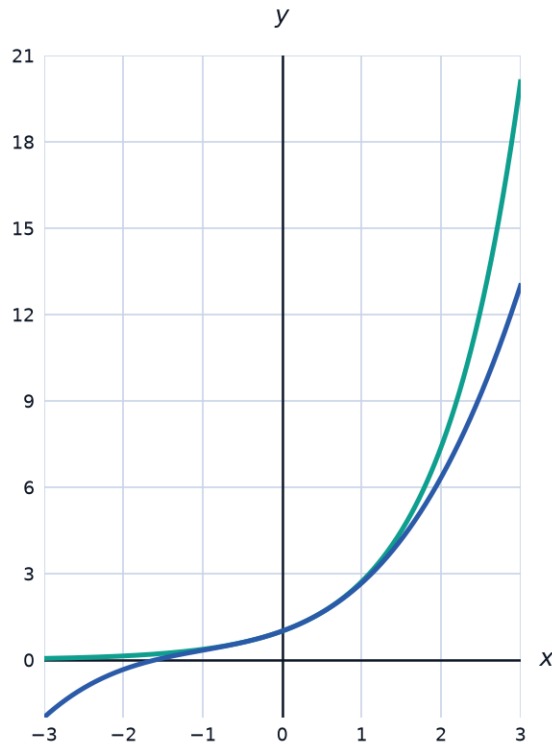


Maclaurin Series & Convergence of Series

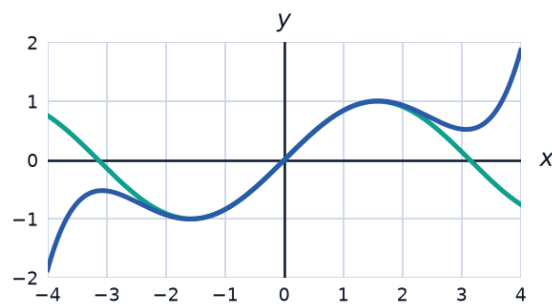


Constructing a Maclaurin series

- 1 Find the first four non-zero terms of the Maclaurin series for $f(x) = e^x$, using $f(x) = f(0) + f'(0)x + (f''(0)/2!)x^2 + (f'''(0)/3!)x^3 + \dots$.



- 2 Find the Maclaurin series for $f(x) = \sin x$ up to the term in x^5 .



Using known series to build new ones

- 3 Using the Maclaurin series for e^x , find the Maclaurin series for e^{-2x} up to the term in x^3 .
- 4 Given the Maclaurin series $\sin x \approx x - x^3/6$ and $\cos x \approx 1 - x^2/2$, find the Maclaurin series for $\tan x$ up to the term in x^3 , using $\tan x = \sin x / (\cos x)$ and polynomial division.

Using a Maclaurin series to approximate a value

- 5 Use the Maclaurin series for e^x up to the x^3 term to approximate $e^{0.1}$, to four decimal places, and compare with the true value $e^{0.1} \approx 1.10517$.
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Tests for convergence of a series

- 6 Use the ratio test to determine whether the series $1/2 + 2/4 + 3/8 + 4/16 + \dots + n/2^n + \dots$ converges.
- 7 Use the ratio test to determine whether the Maclaurin series for e^x , $1 + x + x^2/2! + x^3/3! + \dots + x^n/n! + \dots$, converges for all real x .
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Determining the radius of convergence

- 8 Use the ratio test to find the radius of convergence of the power series $x + x^2/2 + x^3/3 + \dots + x^n/n + \dots$.
- 9 Find the radius of convergence of the series $1 + x + 2x^2 + 6x^3 + \dots + n! x^n + \dots$, and explain what this tells us about where the series is useful.
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Comparing a Maclaurin approximation to the exact function

- 10 Using the Maclaurin series $\cos x \approx 1 - x^2/2 + x^4/24$, approximate $\cos(0.3)$ to four decimal places, and compare with the true value $\cos(0.3) \approx 0.9553$.
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Maclaurin series for $\ln(1+x)$

- 11 Find the Maclaurin series for $f(x) = \ln(1+x)$ up to the term in x^4 , using $f(x) = f(0) + f'(0)x + (f''(0)/2!)x^2 + (f'''(0)/3!)x^3 + (f^{(4)}(0)/4!)x^4$.
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Determining the interval of convergence using the ratio test

- 12 Find the radius of convergence of the Maclaurin series for $\ln(1+x)$, $x - x^2/2 + x^3/3 - \dots + (-1)^{n+1}x^n/n + \dots$, using the ratio test.
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Combining Maclaurin series through multiplication

- 13 Using the Maclaurin series $e^x \approx 1 + x + x^2/2$ and $1/(1-x) \approx 1 + x + x^2$ (valid for small x), find the Maclaurin series for $e^x/(1-x)$ up to the term in x^2 , by multiplying the two series together.
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Using differentiation to find a Maclaurin series

- 14 Given the Maclaurin series $1/(1-x) \approx 1 + x + x^2 + x^3 + \dots$ for $|x| < 1$, differentiate both sides term by term to find the Maclaurin series for $1/(1-x)^2$ up to the term in x^2 .
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Estimating an error bound in a Maclaurin approximation

- 15 The Maclaurin series for $\sin x$ truncated after the x^3 term is $\sin x \approx x - x^3/6$. Use this to approximate $\sin(0.2)$ to four decimal places, and find the absolute error compared to the true value $\sin(0.2) \approx 0.19867$.
- 16 Find the first three non-zero terms of the Maclaurin series for $f(x) = (1+x)^{1/2}$, using the binomial series $(1+x)^n \approx 1 + nx + n(n-1)/2x^2$ with $n = 1/2$.
- 17 Use the Maclaurin series for e^x up to the x^4 term to approximate $\lim_{x \rightarrow 0} (e^x - 1 - x)/x^2$, and check that your answer matches the value obtained using L'Hopital's rule.
- 18 This question has three parts. (a) Find the Maclaurin series for $f(x) = \cos(x^2)$ up to the term in x^8 , by substituting x^2 into the known series $\cos u \approx 1 - u^2/2 + u^4/24$. (b) Use your result to write down an approximation for $\int_0^1 \cos(x^2) dx$ as a sum of simple fractions (this integral has no elementary closed form, so series approximation is genuinely useful here). (c) Evaluate this approximation as a single decimal, to three decimal places.
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Checking endpoints of an interval of convergence

- 19 The Maclaurin series for $\ln(1+x)$ has radius of convergence $R = 1$ (found earlier in this worksheet). Explain why checking the endpoints $x = 1$ and $x = -1$ separately is necessary to determine the full interval of convergence, given that at $x = 1$ the series becomes $1 - 1/2 + 1/3 - 1/4 + \dots$ (the alternating harmonic series, which is known to converge).
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Using a Maclaurin series to evaluate a standard trigonometric limit

- 20 Use the Maclaurin series for $\sin x$ up to the x^3 term to evaluate $\lim_{x \rightarrow 0} (\sin x)/x$.

Identifying the general term of a Maclaurin series

21 Which of the following is the general term of the Maclaurin series for $\cos x$?

- a. $(-1)^n x^{2n}/(2n)!$
- b. $(-1)^n x^n/n!$
- c. $x^{2n}/(2n)!$
- d. $(-1)^{n+1} x^{2n-1}/(2n-1)!$

Comparing the accuracy of two Maclaurin approximations

22 Both $e^x \approx 1 + x + x^2/2$ and $\cos x \approx 1 - x^2/2$ are second-order Maclaurin approximations. Without calculating exact values, explain which approximation is likely more accurate at $x = 1$, referencing the size of the next omitted term in each series.